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Blended wing body configuration for hydrogen-powered aviation

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Abstract

Hydrogen aircraft have the potential to achieve more significant climate impact reductions at a lower cost than aircraft powered by biofuels or other drop-in sustainable aviation fuels. Nevertheless, even as a liquid, hydrogen requires four times more volume than kerosene to store the same energy. Companies and researchers have suggested that the blended wing body configuration is well-suited to hydrogen because it can efficiently store large fuel tanks. However, nobody has quantified this claim, at least publicly. We address this gap by comparing optimized kerosene and hydrogen versions of blended wing body and tube and wing aircraft. Our models predict that with ambitious hydrogen tank technology assumptions, a hydrogen blended wing body has a 3.8% energy penalty compared to a kerosene blended wing body. In contrast, the energy penalty for a tube and wing is 5.1%. These results suggest that the blended wing body configuration is slightly better suited to hydrogen than a tube and wing. However, the conclusion breaks down under less optimistic scenarios where lightweight conformal liquid hydrogen tanks are unavailable. Nonetheless, the energy penalties of switching the two configurations to hydrogen have a similar order of magnitude. Any kerosene blended wing body fuel burn benefit over a tube and wing also applies to hydrogen versions. As a result, kerosene blended wing body technology developments, such as maturing the structural design of the noncircular pressurized passenger cabin, remain valuable for future hydrogen-powered aircraft.

Nomenclature

A_{flapped}	fraction of wing area with flaps	C_l	2D airfoil lift coefficient
BWB	blended wing body	c_{flap}	flap chord to wing chord ratio
b_{slat}	slat span to wing span ratio	c_{slat}	slat chord to wing chord ratio
C_D	drag coefficient	E	Young's modulus
C_d	2D airfoil drag coefficient	f_{stiff}	multiplier to account for stiffeners
C_f	skin friction coefficient	L	length
C_L	lift coefficient	LH_2	liquid hydrogen

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MAC	mean aerodynamic chord	W_{fuel}	fuel weight
MEOP	maximum expected operating pressure	$W_{\text{mission fuel}}$	mission fuel weight (excluding reserve)
MTOW	maximum takeoff weight	W_{payload}	payload weight
OEW	operating empty weight	W_{tank}	hydrogen tank weight
p	pressure	$W_{\text{total fuel}}$	fuel weight including reserve fuel
Re	Reynolds number	β	fraction of boundary layer that is laminar
r	radius	δ	flap deflection angle
T&W	tube and wing	γ	safety factor
TSEC	thrust-specific energy consumption	Λ	quarter chord sweep angle
TSFC	thrust-specific fuel consumption	η_{grav}	tank gravimetric efficiency
t	thickness	ρ	density
t/c	airfoil thickness-to-chord ratio	σ_{yield}	yield stress

1 Introduction

Climate change is driving renewed interest in hydrogen-powered aircraft because they emit no carbon dioxide in flight. These aircraft can decrease the CO₂ equivalent emissions by 50–99% [1, 2] compared to kerosene-powered aircraft. Governments, particularly in Europe, have been motivated by climate change to regulate the commercial aviation industry and fund international programs to investigate the merits of hydrogen-powered aircraft. With the prospect of having to meet stricter regulations, large aerospace companies including Airbus [3], CFM International [4], Pratt & Whitney [5], and Rolls-Royce [6] have announced hydrogen projects.

At ambient temperature and pressure, hydrogen is a gas with far too low a density to fit on an aircraft—its storage would require hundreds of times the volume of the aircraft. To overcome hydrogen’s low volume at ambient conditions, it can be compressed to high pressure or liquefied by cooling it to 20 K. Compressed hydrogen tanks weigh five to ten times more than the hydrogen they contain, while liquid hydrogen tanks weigh the same to twice as much as the hydrogen they store [7]. Thus, liquid hydrogen storage is the most promising solution for long-range aircraft. Liquid hydrogen’s low temperatures require tanks with a low surface area-to-volume ratio to reduce the heat entering the cryogenic fuel. This, in combination with hydrogen’s larger volume requirements, means that fuel can no longer be stored in the wingbox, as kerosene fuel is today. A common solution for tube and wing (T&W) configurations is to store the hydrogen in fuselage tanks [3, 8–12]. Figure 1a shows a popular fuselage packaging solution that allows the tanks to be as close to a sphere as possible because they use the full fuselage radius. However, this solution still increases the aircraft’s weight and drag.

Researchers have found a wide range in the energy consumption difference of hydrogen-powered T&W aircraft compared to kerosene-powered T&Ws, primarily driven by tank weight assumptions [7]. Mukhopadhyaya and Rutherford [2] predict an energy increase of 8–20% for a hydrogen turboprop compared to kerosene and 11–36% for a narrowbody. They assume tank gravimetric efficiencies of 20 and 35%. Maniaci [13] finds that kerosene and hydrogen versions of a 747-sized aircraft use the same amount of energy. Verstraete [14] estimates 10% energy savings for a long-range transport aircraft with optimistic tank weight assumptions.

In recent decades, blended wing body (BWB) designs have matured [15]. Fuel burn reduction estimates range from 15 to 50% compared to T&W aircraft [16, 17]. In collaboration with Cranfield Aerospace and the U.S. Air Force Research Laboratory, NASA and Boeing extensively flight tested scale models of potential BWB configurations [18]. They pushed the flight vehicles to the edges of the flight envelope, particularly in the low-speed flight regimes. A challenge inherent to BWBs is

containing pressure loads in a passenger cabin with a noncircular cross-section. The PRSEUS project, carried out by NASA and Boeing, tested a possible solution of combining the skin, stringers, and frame into a single composite structure [19].

While the T&W configuration has become standard for transport aircraft, hydrogen’s unique storage requirements may incentivize new configurations in ways that kerosene has not. A hydrogen blended wing body (BWB) configuration (Figure 1b) is often suggested as a promising solution because it might efficiently package large hydrogen tanks. Airbus proposed a hydrogen BWB concept in 2020 [20], pointing to its large storage area. For the same reason, JetZero argues that there is a compelling case for a hydrogen-powered BWB [17]. Winnefeld et al. [21] speculate that BWBs enable more flexible hydrogen tank integration than a conventional T&W configuration, which could make them a desirable hydrogen aircraft configuration. Karpuk et al. [22] design kerosene- and hydrogen-powered BWBs and discover challenges efficiently integrating cylindrical tanks. They estimate that the hydrogen BWB uses 27% more energy than the kerosene BWB. However, they do not estimate a similar value for a T&W that could be compared with the BWB results. Guynn et al. [23] study a hydrogen fuel cell BWB, but do not investigate the impact of LH_2 tank integration. Wilod Versprille [24] explores LH_2 tank packaging in a BWB, but does not comment on the impact of tank packaging on the BWB’s performance.

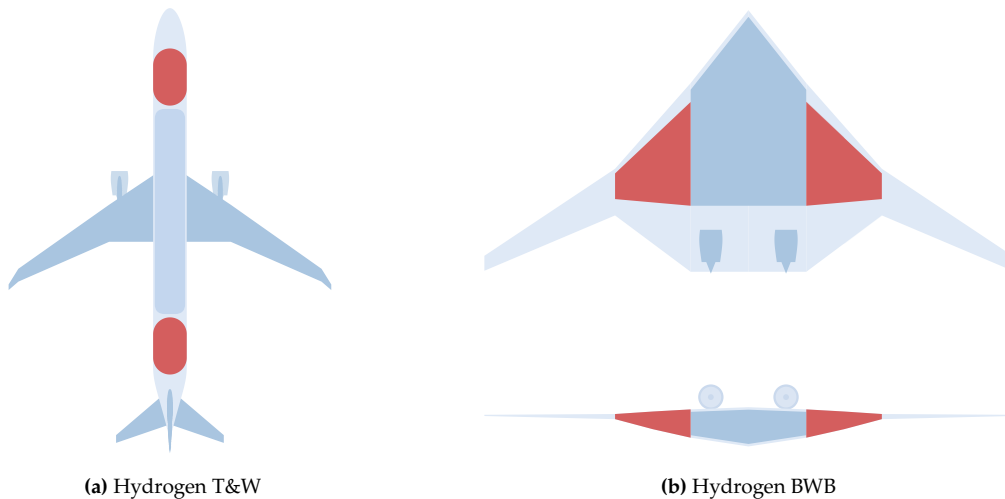


Figure 1: Most hydrogen T&W concepts store hydrogen in the fuselage, which adds weight and drag. The BWB may store the hydrogen more efficiently.

Researchers have yet to quantify the benefit of the BWB configuration compared to T&W for hydrogen-powered aircraft. In this work, we seek to answer that question by optimizing kerosene- and hydrogen-powered versions of each configuration using our conceptual aircraft design tool. For each configuration, we quantify the change in energy consumption, a fuel-agnostic efficiency metric, when the design is adapted from kerosene to hydrogen. By comparing the BWB’s change in energy usage to the T&W’s, we can determine if switching to hydrogen incentivizes the adoption of the BWB configuration.

2 Methodology

The goal of this work is to compare how liquid hydrogen storage requirements affect the energy usage of T&W and BWB airplane configurations. To do this, we optimize four aircraft concepts: a kerosene-powered T&W, a hydrogen-powered T&W, a kerosene-powered BWB, and a hydrogen-

powered BWB. We compare the energy usage of each hydrogen-powered airplane to the kerosene-powered version of the same configuration to investigate how switching to hydrogen fuel affects each configuration's energy consumption. Energy consumption is a natural objective function to answer the research question at hand because it represents the efficiency of each configuration. Other objective functions, such as cost or environmental impact, may result in slightly different aircraft designs [25]. These other objectives should be used if the ultimate goal is to determine which configuration is economically or environmentally superior.

The T&W and BWB configurations employ different modeling approaches that may overpredict or underpredict reality. Therefore, comparing the energy consumption of the hydrogen BWB directly to the kerosene T&W would raise more questions than answers. Higher fidelity models would be necessary for a direct fuel burn comparison between the two configurations, particularly given the limited empirical BWB data. Instead of a direct comparison, we study the relative performance of each configuration's hydrogen-powered version to its kerosene-powered counterpart. This compensates for some of the modeling errors.

The mission requirements are listed in Table 1. The design mission is 5,500 nmi with a payload of 116,000 lbs (a maximum of 420 passengers), approximately the Boeing 787-9's maximum range at its maximum payload [26]. Cruise is at Mach 0.8 and 35,000 ft. The selected cruise altitude is the upper bound of the engine model discussed in Section 2.1.4. The optimal cruise altitudes of the two configurations might differ, but we anticipate that this difference would have a small effect on the conclusions. The mission fuel includes the fuel from the takeoff, climb, cruise, and descent phases. The total fuel required includes the mission fuel plus the fuel for a 200 nmi reserve mission and 30 min loiter. The total fuel also includes an additional 6% to account for variability in engine performance and unreachable fuel [27]. The mission profile is shown in Figure 2.

Table 1: Design requirements

Requirement	Value
Design range	5,500 nmi
Payload weight	116,000 lbs
Takeoff field length	9,000 ft
Maximum passengers	420
Cruise Mach number	0.8
Cruise altitude	35,000 ft
Reserve mission	200 nmi + 30 min loiter

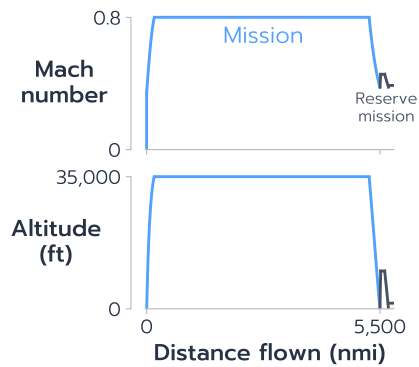


Figure 2: The design mission is 5,500 nmi with a 200 nmi reserve mission.

OpenConcept¹ [28], a conceptual aircraft design toolkit, performs the mission analysis and time integration. OpenConcept is built on OpenMDAO [29] to enable efficient derivative computation. These derivatives allow efficient solution of large optimization problems by harnessing the power of gradient-based optimization [30, Chapters 4–6]. OpenConcept uses numerical integrators to integrate aircraft states, such as fuel burn, over a customizable mission profile. At each numerical integration point, it selects the correct lift coefficient and engine throttle to balance lift, drag, thrust, and weight. The mission analysis is solved simultaneously using a monolithic Newton solver that leverages the built-in analytic derivatives. The toolkit can handle complexity ranging from models with a simple parabolic drag polar and propulsion model to models using vortex lattice-based aerodynamics, hybrid-electric engine maps, and unsteady modeling of a battery and electric motor thermal management system. We do not consider fuel cells or other electrified propulsion in this work.

The optimizer performs the aircraft sizing, such as determining the appropriate wing area and rated thrust, subject to the mission requirements. Simultaneously, it determines the optimal values of all other design parameters to minimize energy consumption. This approach circumvents the need for manual aircraft sizing prior to the optimization. It is also superior because the sequential approach leads to suboptimal designs [30, Section 13.1]. The remaining portions of this section describe the methods used to model the T&W and BWB configurations, along with a validation case of the models.

2.1 Tube and wing

An OpenConcept airplane model must compute weight, drag, and thrust as a function of lift coefficient, throttle, and flight conditions. Empirical relations estimate the T&W’s weight. The drag prediction combines empirical textbook drag buildup methods and a physics-based vortex lattice code. A one-dimensional engine cycle model provides the thrust and fuel flow. The initial parameters are based on the Boeing 787-9.

2.1.1 Geometry, sizing, and hydrogen tank integration

For the T&W concept, the fuselage is extended by the total length of the two hydrogen tanks to accommodate them without sacrificing passenger capacity. The stretched fuselage length is then used for all calculations. To maintain an appropriate rotation angle and capture trends in landing gear weight, the landing gear height is scaled by the ratio of the extended fuselage length to the original fuselage length.

The geometric design variables are shown in Figure 3. The wing area uniformly scales the entire wing. This links the wing area to the span, which may incentivize increasing the area to reduce induced drag. Wing twist controls the twist at the wing break and two outboard sections, linearly interpolating in between. The optimizer also controls the tank radius and length. Both tanks have the same geometry.

Planform variables are excluded from the optimization. The optimal aspect ratio is a tradeoff between aerodynamic performance and structural weight. Sweep is also a tradeoff between structural weight and wave drag. More detailed physics-based models are necessary to capture these tradeoffs properly. Wing twist design variables are included in the optimization because the 787’s twist distribution is not readily available, and achieving reasonable twist distributions does not rely as heavily on an accurate structural model. By including twist, the optimizer can achieve a more realistic lift distribution than a flat wing would provide.

The vertical and horizontal tail areas are computed using tail volume coefficients [27, Equations 6.26 and 6.27]. The tail lever arm is assumed to be half the fuselage length. Other geometry parameters

¹<https://github.com/mdolab/openconcept>

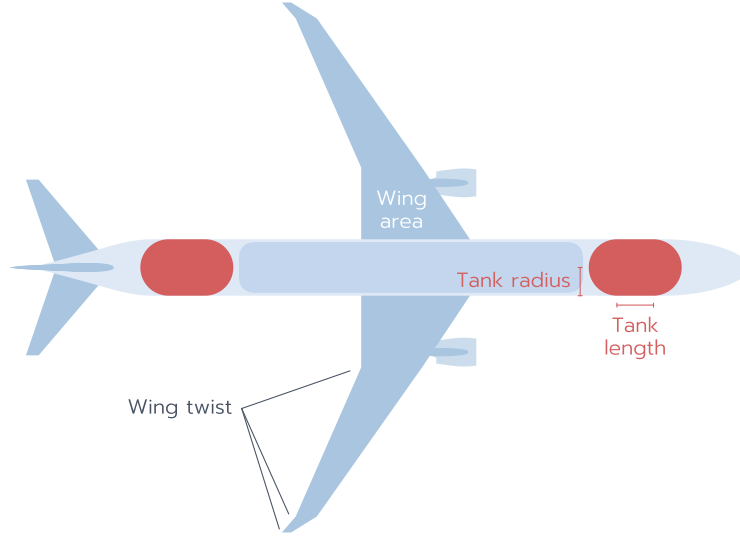


Figure 3: T&W geometry parameterization.

for the T&W concepts are taken from the Boeing 787-9.

2.1.2 Weight estimation

The in-flight weight is the sum of the fuel weight, payload weight, and operating empty weight. The fuel weight is computed by numerically integrating the fuel flow from the propulsion model and subtracting it from the initial fuel weight. The operating empty weight of the kerosene T&W aircraft is computed using empirical correlations. The wing, horizontal tail, vertical tail, fuselage, and landing gear weights are estimated by Raymer [27, Equations 15.25–15.30]. The maximum landing weight, an input to the landing gear weight, is computed by subtracting two-thirds of the fuel weight from the maximum takeoff weight. Baseline engine, thrust reverser, engine control, and fuel system weights are computed using Wells et al. [31, Equations 76, 86, 87, and 92]. Correlations from Roskam [32, Equations 5.37 and 6.27] predict nacelle and engine starter weight. Furnishing and other equipment weights are also computed with Roskam [32, Equations 7.6, 7.15, 7.23, 7.29, 7.35, 7.40, and 7.44]. The structural weights, including the wing, tails, fuselage, landing gear, and nacelles, are multiplied by 1.2. The final operating empty weight is multiplied by an additional 1.15. These multipliers account for miscellaneous items that were not considered and compensate for some of the empirical models' tendency to underestimate real-world values. With these factors held constant, the operating empty weight model predicts reasonable weight values for a wide range of transport aircraft.

A physics-based model estimates the cryogenic tank weight for the hydrogen T&W concept. The model assumes a vacuum-insulated aluminum tank construction because it is a promising tank design for future hydrogen aircraft [7]. The tank is cylindrical with two hemispherical endcaps. The vacuum-insulated tank has an inner shell that contains the pressure of the LH_2 inside the tank and separates the tank interior from the vacuum region. The vacuum region is enclosed by an outer shell that resists the compression of the external atmospheric pressure. The inner shell's thickness is sized by the hoop stress using

$$t_{\text{in}} = \frac{p_{\text{in}} r_{\text{in}} \gamma_{\text{in}}}{\sigma_{\text{yield, in}}}, \quad (1)$$

since the pressure of the tank's interior is greater than the pressure in the vacuum region. The maximum expected operating pressure (MEOP) of the tank is 2 bar, which is in the range of reasonable tank operating pressures [9, 33]. The outer shell is under compression because the atmospheric

pressure on its exterior is greater than the pressure of the vacuum region. Thus, its thickness is sized to support buckling loads [33]. We use elastic buckling relationships for spherical and cylindrical shells under external pressure from Budynas and Sadegh [34, Table 15.2]:

$$t_{\text{out, sph}} = r_{\text{out}} \sqrt{\frac{p_{\text{out}} \gamma_{\text{out}}}{0.365E}}, \text{ and} \quad (2)$$

$$t_{\text{out, cyl}} = \left(\frac{p_{\text{out}} \gamma_{\text{out}} r_{\text{out}}^{1.5} L}{0.92E} \right)^{0.4}. \quad (3)$$

The outer thickness is the maximum of the thicknesses required for the spherical and cylindrical geometries multiplied by a factor f_{stiff} to account for the addition of stiffeners, which Sullivan et al. [33] show can substantially decrease the outer shell's weight. The parameters used for these calculations are listed in Table 2. The tank weight is then computed for each shell by multiplying its thickness by its surface area and material density. The final tank weight is multiplied by 1.1 to account for supports, plumbing, and other necessary equipment. For the hydrogen concepts, the operating empty weight is computed using the same models as the kerosene-powered concept, and the hydrogen tank weight is added to the result.

Table 2: Hydrogen tank structural parameters

Parameter	Value	Notes
p_{in}	200,000 Pa	MEOP of the internal tank
p_{out}	101,325 Pa	Representative maximum external pressure
γ_{in}	1.5	
γ_{out}	2.0	
$\sigma_{\text{yield, in}}$	413.7 MPa	Al 2014-T6 yield stress [33, Table IV]
E	80 GPa	LiAl 2090 Young's modulus [33, Table XIII]
ρ_{in}	2,796 kg/m ³	Al 2014-T6 density [33, Table IV]
ρ_{out}	2,699 kg/m ³	LiAl 2090 density [33, Table XIII]
f_{stiff}	0.8	

The wing geometry is not suited to store the insulated cryogenic tanks for hydrogen aircraft [7]. Removing fuel from the wings eliminates the fuel load alleviation, which increases the weight of the wing structure. To account for this, we compare the wing weight relief factors provided by Jenkinson et al. [35] with and without the fuel weight load alleviation for a representative aircraft. Based on this estimate, we include a 20% wing weight penalty for the hydrogen concepts. The 20% weight penalty is a similar order to the penalty computed by performing steady aerostructural optimization with and without the fuel load alleviation [36].

2.1.3 Aerodynamics

We use a vortex lattice code to model the wing aerodynamics and textbook methods to estimate the drag of other components. The vortex lattice implementation is OpenAeroStruct [37], which is written using the OpenMDAO framework and provides analytic derivatives. OpenAeroStruct estimates parasitic drag with a flat-plate skin friction estimate and a form factor correction. It estimates wave drag with relations based on the Korn equation [38]. It is integrated into OpenConcept using a surrogate model [39]. Whenever the wing geometry changes, OpenConcept's surrogate model reruns OpenAeroStruct to generate a new surrogate model. The OpenAeroStruct mesh for the T&W concept is shown in Figure 4.

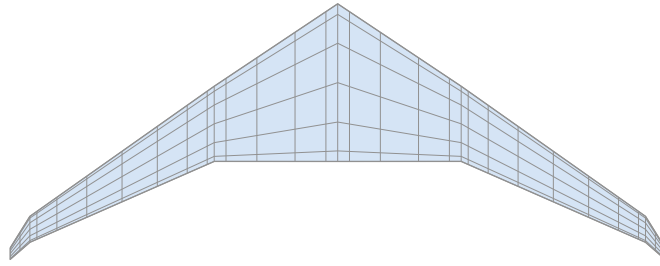


Figure 4: OpenAeroStruct mesh used for the T&W configuration.

An empirical drag buildup² is used to account for the drag of components other than the wing. The fuselage, horizontal tail, vertical tail, and nacelles use empirical form factor relations from Torenbeek [40] and interference factors from Raymer [27], listed in the appendix. Skin friction coefficients are then computed for each component based on its characteristic length and flight conditions, also using a process described in the appendix. The skin friction coefficient is a weighted sum of laminar (Blasius) and turbulent (explicit fit of Spalding [41]) estimates. The drag coefficient of each component is computed by multiplying the form factor by the interference factor and skin friction coefficient. The resulting drag coefficients are renormalized by the wing planform area and added to the total drag coefficient. The flaps and landing gear are extended during takeoff, which adds drag. We add 200 drag counts to account for the extended landing gear [27]. Flap drag is computed using a curve fit of Roskam [42] estimates for Fowler flaps, listed in the appendix.

The maximum lift coefficient in the takeoff configuration is needed to estimate the takeoff field length. The clean maximum lift coefficient is based on an empirical relation from Raymer [27, Equation 12.15] between the two-dimensional airfoil maximum lift coefficient and the sweep angle. The clean two-dimensional airfoil maximum lift coefficient is 1.6. Extending the flaps and slats increases the maximum lift coefficient, which is modeled using correlations for Fowler flaps and slats from Roskam [42], listed in the appendix. Takeoff flap deflection is assumed to be 10 degrees.

2.1.4 Propulsion

We use a CFM56 engine model adapted from Adler et al. [43], which uses a surrogate model of the pyCycle [44] CFM56 engine model. To match more modern engines, we assume a 20% TSFC improvement. We have modified it to use hydrogen by assuming the same thrust-specific energy consumption (TSEC) as the kerosene-powered version, which offers a reasonable estimate of hydrogen fuel consumption [2, 7, 13]. The engine's thrust and fuel flow are linearly scaled to match the desired engine rating while maintaining the same TSFC.

Using liquid hydrogen enables a range of potential technologies to improve thermal efficiency. For example, Boggia and Jackson [45] propose using the cryogenic LH₂ as a heat sink before the compressor and upstream of the turbine. They find that this enables single-digit percent improvements in TSFC. Recovery and injection of the water vapor produced by the hydrogen combustion may also improve engine efficiency [5, 46]. The sensitivity to the hydrogen engine's fuel consumption is investigated in Section 4.

²The parasitic drag estimations are synthesized here: <https://openvsp.org/wiki/doku.php?id=parasitedrag> (accessed April 24, 2023)

2.1.5 Optimization problem

The T&W optimization, listed in Table 3, minimizes the mission fuel burn by varying the wing area, turbofan rated thrust, hydrogen tank geometry, and wing twist. The wing’s planform geometry is based on the Boeing 787’s wing to represent modern sweep, aspect ratio, and taper ratio values. The thickness-to-chord ratio of the wing ranges from 13.4% at the root to 10% at the tip. Twist is controlled at three spanwise stations, and the root is locked to prevent rigid body rotations of the wing. The hydrogen tank radius is bounded to ensure it fits within the fuselage. For the hydrogen T&W concept, we assume two equally-sized hydrogen tanks at the front and rear of the fuselage. This offers more center of gravity control than a single tank at the back of the fuselage.

The optimization is constrained so that the hydrogen tank is large enough to store the fuel required for the mission. The tank is not allowed to exceed a fill level of 95% because there must always be some room for the LH_2 to boil off to avoid drastic increases in tank pressure [7, 33]. The takeoff field length is computed using the method described by Brelje and Martins [28]. It is constrained to be less than 9,000 ft, in line with the Boeing 787-9 on a standard day at sea level and MTOW [26]. Finally, the throttle must remain between 1 and 100% throughout the mission, which prevents the optimizer from selecting an engine that cannot meet the thrust requirements. This same throttle constraint could be accomplished by directly bounding the throttle in OpenConcept’s Newton solver. However, allowing OpenConcept’s Newton solver to select throttle values up to 120% and then constraining throttle to 100% in the optimization problem avoids analysis failures that slow the optimization.

We also use the optimizer to solve for the maximum takeoff weight by adding a constraint that the takeoff weight must be at least the sum of empty weight, payload weight, and fuel weight. Doing this removes an internal solver feedback loop, which reduces the overall optimization time and improves the Newton solver’s robustness. Following the convention of past OpenConcept work [28, 47], we impose the constraint as an inequality. This constraint is always active, so in practice, this formulation is equivalent to an equality constraint.

Table 3: Optimization problem for T&W aircraft

minimize	$W_{\text{mission fuel}}$
by varying	Engine rated thrust Wing area Hydrogen tank radius ³ Hydrogen tank length ⁴ Wing twist MTOW
subject to	Takeoff field length $\leq 9,000$ ft $1\% \leq \text{throttle} \leq 100\%$ $0\% \leq \text{hydrogen tank fill level} \leq 95\%$ ⁵ $\text{MTOW} - (\text{OEW} + W_{\text{payload}} + W_{\text{total fuel}}) \geq 0$

We use SNOPT [48] through the pyOptSparse [49] interface. All optimizations are converged to an optimality tolerance of 10^{-6} and feasibility tolerance of 10^{-8} , decreasing optimality from the baseline by roughly five orders. The optimizations take 5–10 min on a desktop computer with an AMD Ryzen 9 5950X.

³Tank radius is included only for the hydrogen T&W

⁴Tank length is included only for the hydrogen T&W

⁵Fill level constraint is included only for the hydrogen T&W

2.2 Blended wing body

To model the BWB, we divide the geometry into three regions, shown in Figure 5. The center region is called the *centerbody*, which contains a pressurized region for the passengers. Just outboard of the centerbody is the *blending region*, which stores the fuel for the hydrogen concepts. Finally, there is a higher aspect ratio wing outboard of the blending region. The combination of the furthest outboard wing and the blending region is called the *outerwing*. The outerwing stores fuel for the kerosene concept. The reason for storing hydrogen in the blending region rather than the whole outerwing is because a tank in the thinner outboard wing would have too high a surface area-to-volume ratio to accommodate the cryogenic LH₂ efficiently [7]. The absolute thickness of the blending region is greater, so it is possible to package tanks well suited to LH₂ storage. Unless otherwise specified in this section, the BWB concepts use the same models as the T&W concepts.

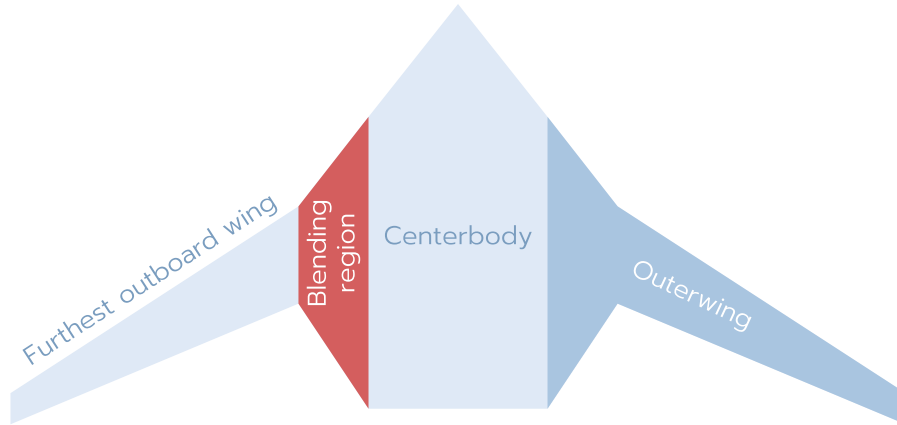


Figure 5: The BWB geometry is defined by three linearly-varying regions.

2.2.1 Geometry, sizing, and hydrogen tank integration

The baseline BWB geometry is designed to provide a comparable cabin floor area to the Boeing 787-9's cabin. The initial centerbody and outerwing planform shapes are chosen to align with designs from Boeing [15], MIT [50, 51], JetZero⁶, and other similar concepts. Similar to a number of other BWB concepts [15, 51, 52], we do not use inboard vertical tails. Advanced structural and propulsion technologies, such as boundary layer ingesting engines, are also omitted in this study. The configuration uses two turbofans, the trend with modern aircraft because it enables higher efficiencies than using three or four engines. We assume the pressurized passenger cabin extends from the leading edge to the rear spar, located at 65% chord in the outerwing and extending straight through the centerbody. The front spar is at 10% chord in the outerwing.

We initially attempted to package a single cylindrical LH₂ tank with hemispherical endcaps into each blending region, shown in Figure 6. With our parameterization, the blending region could not be made big enough to fit sufficiently large tanks. A BWB with cylindrical LH₂ tanks may be made feasible by using a different geometry parameterization or packaging multiple cylindrical tanks in each blending region [22]. However, Karpuk et al. [22] run into several challenges integrating multiple cylindrical tanks into the blending region. They find that the only feasible solution to achieve enough tank volume is to use a far longer and more slender cabin than the kerosene BWB, which has negative aerodynamic efficiency, safety, and landing gear integration implications. As the number of cylindrical tanks in each blending region increases, the shape of the storage region approaches that of

⁶<https://www.jetzero.aero/> (accessed April 24, 2023)

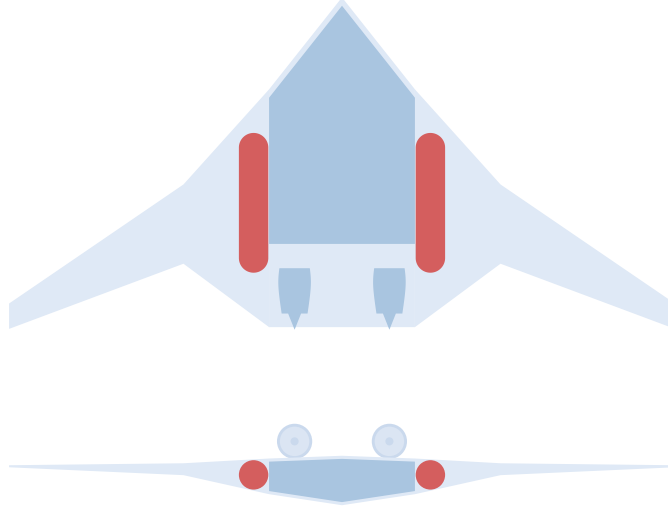


Figure 6: A cylindrical tank in each blending region does not have sufficient volume. The tanks in this figure contain only half of the hydrogen needed. A more refined geometry parameterization could potentially make this configuration feasible.

the conformal tank. The storage density would be less because of the lower packing efficiency of the cylinders. The gravimetric efficiency may be higher or lower, depending on the tradeoff between using an optimal pressure vessel shape and designing a tank that best takes advantage of the available space. These effects result in drag and weight penalties for the cylindrical tank approach. We investigate the sensitivity to the tanks' gravimetric efficiency in Section 4. A detailed tank packaging integration study, similar to the one by Brelje and Martins [53], would be useful to quantify the weight and drag penalties of using cylindrical LH₂ tanks compared to conformal ones.

Because cylindrical LH₂ tanks do not fit, we consider conformal tanks that fill the space in the blending region between the front and rear spars (Figure 7). The volume of these tanks is computed by integrating the thickness from 10–65% chord of a NASA SC(2)-0412 airfoil scaled to match the thickness-to-chord ratios of the inner and outer blending region sections. The tanks are formed by linearly lofting between the two airfoil sections and integrating to compute the total tank volume.

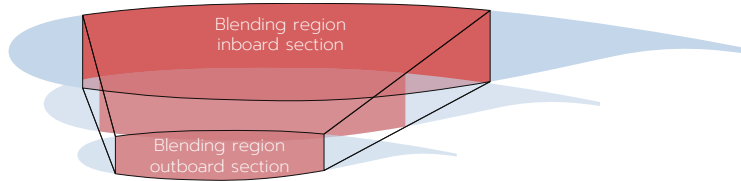


Figure 7: The conformal tank volume is computed by integrating the cross-sectional area at the inboard and outboard blending region sections and linearly lofting them.

The BWB geometry parameterization must give the optimizer enough freedom to shape the blending region to store the hydrogen tanks and capture the critical packaging tradeoffs.⁷ Physical considerations such as structural design and significant three-dimensional flow field effects are missing from the low-fidelity model. Thus, the parameterization and geometry constraints must be restrictive enough to obtain reasonable designs.

The parameterization for this work, shown in Figure 8, locks the centerbody planform geometry

⁷For example, if the optimizer can adjust the blending region width but not the width of the furthest outboard wing, it widens the blending region to increase the overall span. This makes the blending region artificially large and reduces the apparent penalty of packaging hydrogen tanks.

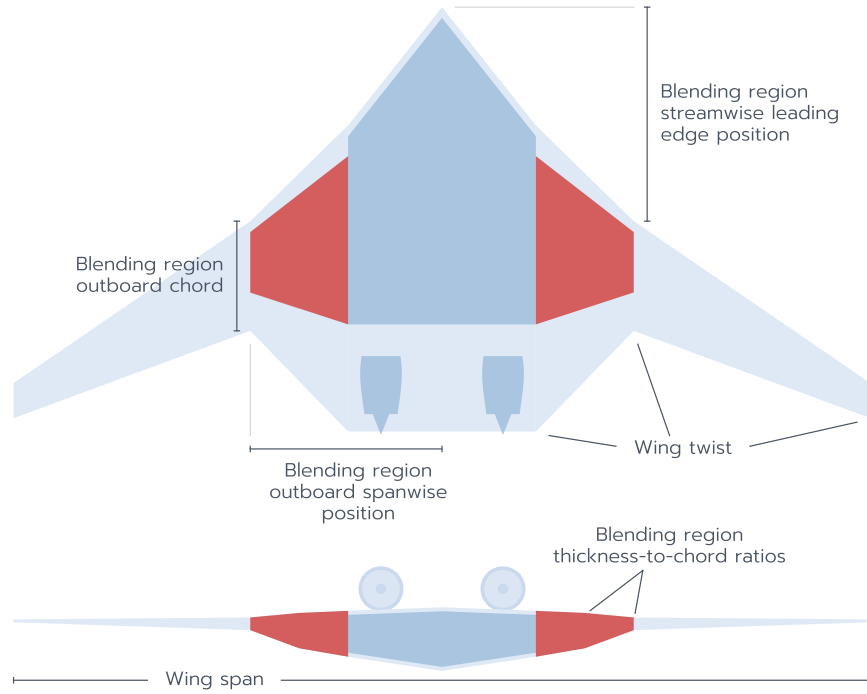


Figure 8: The BWB geometry parameterization focuses on design freedom around the blending region.

and thickness-to-chord ratio (set to a constant 15%). The optimizer can vary the thickness-to-chord ratio, chord, spanwise position, and streamwise position of the outboard blending region section. It can also adjust the thickness-to-chord ratio of the section in the middle of the blending region. Lastly, it can change the spanwise position of the outermost wing section. Similarly to the T&W wing geometry parameterization discussed in Section 2.1.1, the planform shape of the BWB's outerwing is a delicate tradeoff between aerodynamic performance and structural weight. The structural model does not have sufficient fidelity to quantify these tradeoffs properly, so the furthest outboard wing region's quarter chord sweep angle and taper ratio are held constant, as is the wing tip's thickness-to-chord ratio. Like the T&W, twist design variables are included so the optimizer can obtain a more realistic lift distribution. Unlike the T&W, the furthest outboard wing's aspect ratio cannot be held constant because then the optimizer increases the blending region outboard chord as a way of increasing the overall span to reduce induced drag. This results in unrealistically large blending regions that deceptively lessen the hydrogen tank integration penalty. Instead, an upper bound on the furthest outboard wing's aspect ratio is imposed to prevent the optimizer from selecting unrealistically slender designs. The streamwise position of the quarter mean aerodynamic chord is constrained to account for structural and stability considerations.

2.2.2 Weight estimation

The structural weight model is the only difference in weight estimation from the T&W. The fuselage, horizontal tail, and vertical tail are replaced by the centerbody, which is divided into two regions: the pressurized cabin and the unpressurized portion aft of the cabin. The weight of both of these regions is estimated using empirical correlations derived from finite element analyses by Bradley [54, Section 4.2]. The outerwing weight is computed using the same wing weight estimate as for the T&W.

Storing the hydrogen in one cylindrical tank with hemispherical endcaps in each blending region

is not feasible (see Section 2.2.1). Thus, we use conformal tanks that fill the region between the front and rear wing spars. Developing a physics-based structural model of a vacuum-insulated conformal LH₂ tank is difficult at low fidelity, so we assume a similar baseline gravimetric efficiency as the physics-based cylindrical tank model of 60% and investigate the sensitivity of the results to this value in Section 4. Gravimetric efficiency represents the fraction of the tank’s total weight taken up by fuel when filled. It is written as

$$\eta_{\text{grav}} = \frac{W_{\text{fuel}}}{W_{\text{tank}} + W_{\text{fuel}}}, \quad (4)$$

where W_{fuel} is the maximum fuel weight the tank can hold and W_{tank} is the weight of the tank structure. A gravimetric efficiency of 60% is predicted to be feasible for vacuum-insulated cylindrical or spherical tanks [55], but is a rather ambitious target for conformal tanks. For example, Collins Aerospace is targeting a gravimetric efficiency of 35% for a variable section box-shaped LH₂ tank [56]. Realistic gravimetric efficiencies for large conformal LH₂ tanks that would fit in the blending region are unavailable. Advanced technologies, such as tank integration into the primary aircraft structure, could help reduce the effective gravimetric efficiency. Given the lack of data, we assume a similar efficiency to the T&W’s tanks and investigate the sensitivity of the results to this assumption in Figure 16.

2.2.3 Aerodynamics

Unlike a T&W aircraft, a BWB’s aerodynamics are dictated almost entirely by the wing. However, the wing has a significantly different shape than conventional wings, which are often nearly trapezoidal. OpenAeroStruct provides a cheap aerodynamic model to capture the planform geometry changes of a BWB’s unconventional wing shape. The mesh (Figure 9) estimates the lift and drag with errors of 0.1% and 1%, respectively, compared to a mesh with 20 times the number of panels. The parasitic drag of the BWB’s lifting surface (i.e., outerwing and centerbody) is computed using OpenAeroStruct’s flat-plate estimate with a form factor correction [37]. The wave drag estimation is based only on the outerwing geometry, similar to the approach taken by Kays [50], rather than OpenAeroStruct’s default of using the whole wing. Besides the wing drag, the nacelles are the only other parasitic drag source modeled in flight using the same approach as the T&W. Because we do not account for pitching moment trim in the planform and twist optimization, an extra 5% drag penalty is added. This is roughly equal to the difference in drag between BWBs optimized with and without a pitching moment constraint found by Lyu and Martins [57].

On takeoff, the landing gear drag is added. The landing gear drag coefficient of 0.02 is halved because it is now normalized by roughly twice the planform area. Trailing edge flaps are excluded

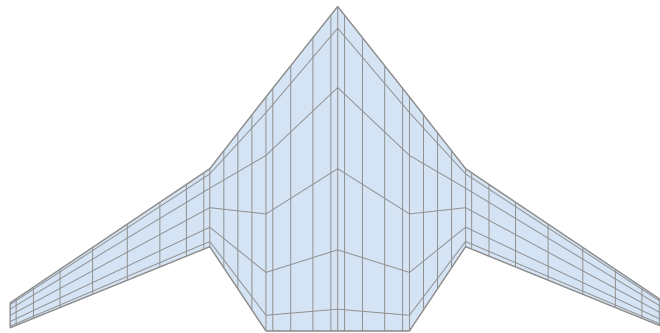


Figure 9: OpenAeroStruct mesh used for the BWB configuration (initial planform geometry).

because the BWB has no tail to trim the pitching moments they produce [15]. The BWB also has a lower wing loading than conventional aircraft because of the additional lifting area the centerbody provides, which means it has adequate high lift performance and flaps are unnecessary [57].

The maximum lift coefficient for the BWB is estimated using the critical section method. This approach determines the highest angle of attack where all sectional lift coefficients are less than or equal to their maximum two-dimensional airfoil lift coefficient. The wing's lift coefficient at that angle of attack is taken to be the maximum lift coefficient. High-lift design for BWBs is still an active area of research, so we assume a maximum sectional lift coefficient value of 1.7. The angle of attack is solved using OpenMDAO's nonlinear Schur complement solver [58], which circumvents the need to assemble and solve a linear system that contains the entire OpenAeroStruct vortex lattice model. Using the Schur solver decreases the run time of the maximum lift coefficient model by a factor of 20 compared to using a Newton solver.

2.2.4 Propulsion

The BWB uses the same engine model as the T&W configuration with the same methods of adapting to hydrogen fuel and variable engine ratings.

2.2.5 Optimization problem

The BWB optimization problem, listed in Table 4, is the same as the one for the T&W aircraft, except that it replaces the T&W's geometric design variables with the corresponding BWB parameters, described in Section 2.2.1. The lower bound on blending region outboard thickness-to-chord ratio is 11%. The blending region outboard spanwise position is not allowed to encroach too closely on the outboard section of the centerbody (this constraint is active for the optimizations without a hydrogen tank fill level constraint). The wing span is limited to be no greater than the baseline.

Table 4: Optimization problem for BWB aircraft

minimize	$W_{\text{mission fuel}}$
by varying	Engine rated thrust Blending region mid thickness-to-chord ratio Blending region outboard thickness-to-chord ratio Blending region outboard chord Blending region outboard spanwise position Blending region outboard streamwise leading edge position Wing span Wing twist MTOW
subject to	Takeoff field length $\leq 9,000$ ft $1\% \leq \text{throttle} \leq 100\%$ $0\% \leq \text{hydrogen tank fill level} \leq 95\%^8$ Change in streamwise quarter MAC position = 0 Furthest outboard wing aspect ratio ≤ 10 $\text{MTOW} - (\text{OEW} + W_{\text{payload}} + W_{\text{total fuel}}) \geq 0$

⁸Fill level constraint is included only for the hydrogen BWB

2.3 Validation

We validate our approach by modeling a Boeing 787-9 flying a 5,250 nmi mission with maximum payload and comparing to the true values [26]. This mission corresponds to the 787-9’s maximum range at maximum payload. Because we do not know the twist of the 787, we obtain the values for validation by performing a twist-only optimization that minimizes fuel burn. The validation results are shown in Table 5.

Table 5: Boeing 787-9 validation on a 5,250 nmi at maximum payload.

Parameter	Boeing 787-9 [26]	OpenConcept	OpenConcept error
MTOW (lb)	560,000	551,155	−1.6%
OEI (lb)	280,000	281,739	0.6%
Total fuel (lb)	160,000	161,475	0.9%
Takeoff field length (ft)	9,300–9,900 ⁹	7,938	−14.6 to −19.8%

3 Results

In this section, we investigate the results of the optimizations. We first analyze the two T&W aircraft and explain key differences in the optimized designs. Then, we do the same for the two BWB aircraft. Finally, we study how each configuration adapts to hydrogen fuel by comparing each configuration’s change in energy usage from the kerosene to the hydrogen version. Detailed numerical results are listed in Tables 7, 8, 9, 10, and 11 in the appendix.

The optimized T&W aircraft are shown in Figure 10. Packaging the hydrogen tanks results in a longer fuselage, which increases the fuselage drag in cruise by 24%. The stretched fuselage has a length-to-diameter ratio similar to the Boeing 757-300’s fuselage. Lengthening the fuselage increases the moment arm of the tail, which enables smaller horizontal and vertical stabilizers.

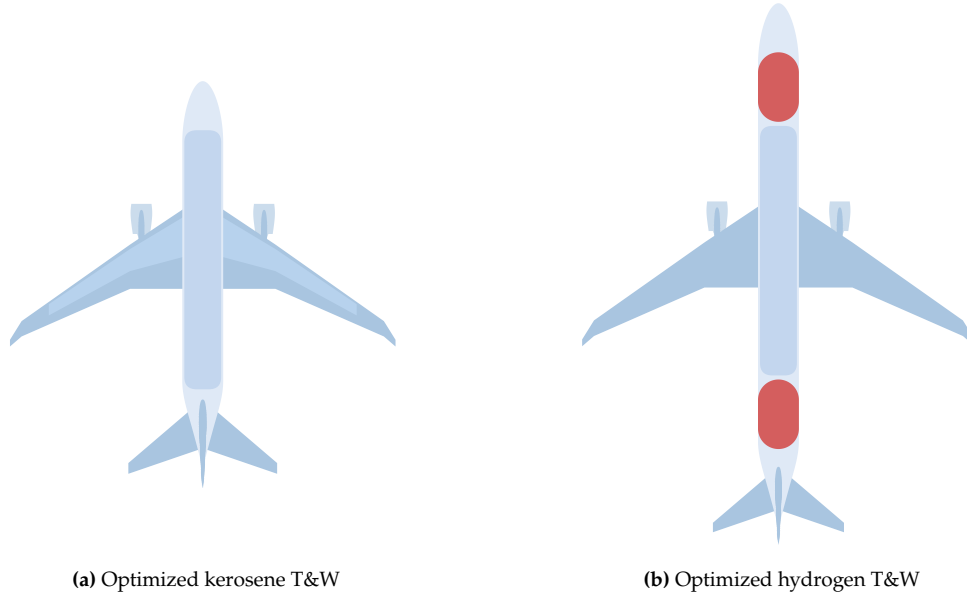


Figure 10: Compared to the optimized kerosene T&W, the hydrogen T&W has a 4% larger wing area, 4% less powerful engines, and a 36% longer fuselage to accommodate the tanks.

⁹Takeoff field length varies with different engine options

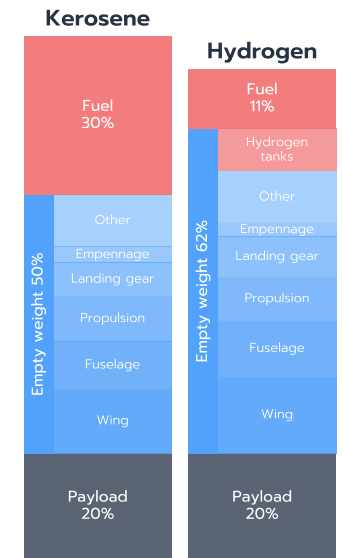


Figure 11: The optimized hydrogen T&W has a lower MTOW but greater OEW than the kerosene version.

The longer fuselage and addition of the hydrogen tanks increase the operating empty weight of the hydrogen T&W compared to the kerosene version. The longer fuselage also requires taller landing gear to achieve the same rotation angles, which increases the landing gear weight by 22%. Despite the greater empty weight, the hydrogen-powered concept's MTOW is 7% less than the kerosene concept's (Figure 11). This is because hydrogen carries three times the energy of kerosene in the same weight, so the fuel weight is reduced by a factor of three. The lighter takeoff weight allows the optimizer to shrink the engines. The lower MTOW also means that the maximum lift force the wing must produce is lower. This reduced lift force slightly offsets the 20% wing weight penalty from the lack of fuel load alleviation, but the wing is still 19% heavier than the kerosene aircraft's wing.

Figure 12 shows the optimized BWB aircraft. The optimizer collapses the blending region of the kerosene-powered BWB as much as the design variable bounds allow. This change shrinks the wetted area, reducing drag. The optimizer also lowers the thickness-to-chord ratio of the blending region's mid and outboard sections to the lower bound of 11%. The span extends to the design variable's upper bound to reduce induced drag. Finally, the chord of the blending region's outboard section shrinks, further reducing wetted area and drag until the takeoff field length constraint prevents further reduction. A smaller wing would not be able to provide enough lift to take off in the required distance.

The hydrogen BWB takes on a different shape than the kerosene BWB because the blending regions must be large enough to store the hydrogen. Due to the larger blending regions, the wetted area is 8% greater than the kerosene BWB's. Despite its starkly different shape and greater planform area than the kerosene BWB, the hydrogen BWB achieves a 9% reduction in MTOW compared to the kerosene BWB (Figure 13). The hydrogen BWB's wing is 12% heavier than the kerosene BWB's. Some of the 20% wing weight penalty from the lack of fuel load alleviation is compensated by the smaller aspect and taper ratios, along with the lower lift force required because of the hydrogen BWB's reduced takeoff weight.

This work aims to determine whether T&W aircraft or BWBs more efficiently adapt to hydrogen aircraft requirements. We measure the efficiency of converting a configuration to hydrogen by comparing the energy usage of the hydrogen and kerosene versions of each configuration. A lower ratio of the hydrogen aircraft to kerosene aircraft's energy consumption indicates a concept that is better suited to hydrogen storage. The result does not determine which configuration uses less energy; it

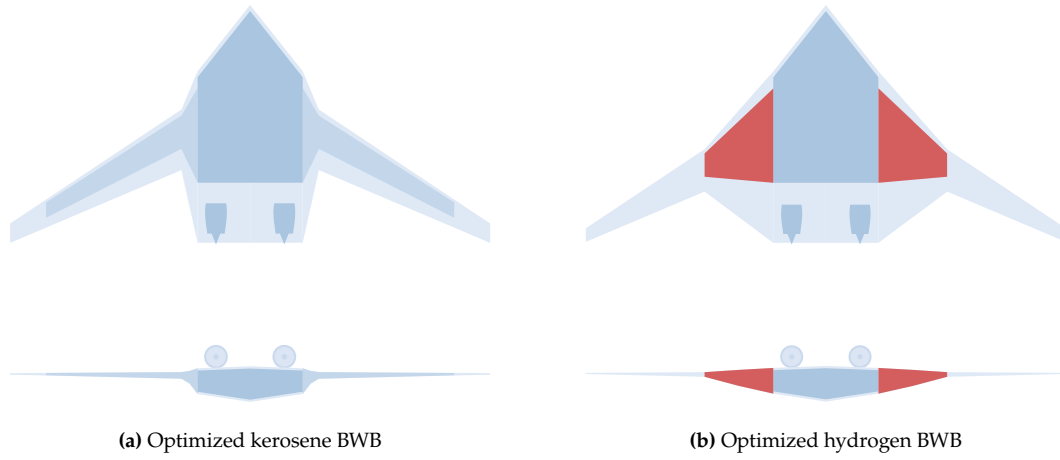


Figure 12: The kerosene BWB collapses the blending region to decrease wetted area, while the hydrogen version widens the blending region to store enough fuel.

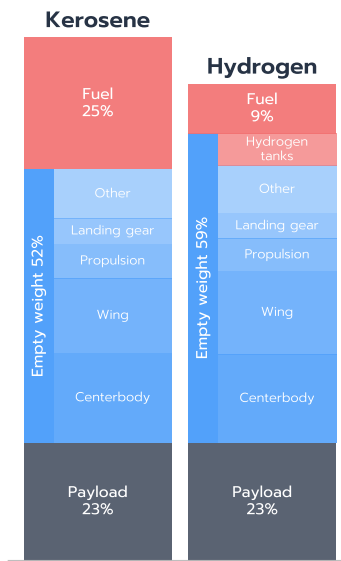


Figure 13: The optimized BWB weight breakdowns show similar trends to the optimized T&W aircraft.

only describes whether storing hydrogen tilts the scales in favor of one of the configurations. This is because we compare each hydrogen aircraft only to the kerosene aircraft with the same configuration. We do not compare, for example, the hydrogen BWB's energy usage directly to the hydrogen T&W's.

Figure 14 shows the difference in energy usage between the hydrogen and kerosene T&W aircraft, while Figure 15 shows the same plot for the BWB. The conversion from kerosene to hydrogen is broken into three steps to offer additional insights into the source of the differences between the two configurations. The design is optimized at each step with the appropriate design variables and constraints. The differences between the steps are the following:

1. **Switch to hydrogen:** Account for the higher specific energy of hydrogen, which results in a 2.8 times lower TSFC. Because there is no longer kerosene stored in the wings, include the 20% wing weight penalty.
2. **Include tank weight:** Include the hydrogen tank weight in the empty weight calculation. For the T&W, this requires adding tank radius and length design variables along with the fill level constraint to the optimization. However, the fuselage is not lengthened to account for the tank

packaging. For the BWB, the tank weight calculation does not require knowledge of the tank geometry, so the fill level constraint is still left out of the optimization problem.

3. **Include tank integration:** Package the hydrogen tank into the airframe. For the T&W, stretch the fuselage to fit the tanks. Stretching the T&W's fuselage also involves lengthening its landing gear. For the BWB, add the fill level constraint to the optimization, which forces the optimizer to make the blending region large enough to store the required hydrogen.

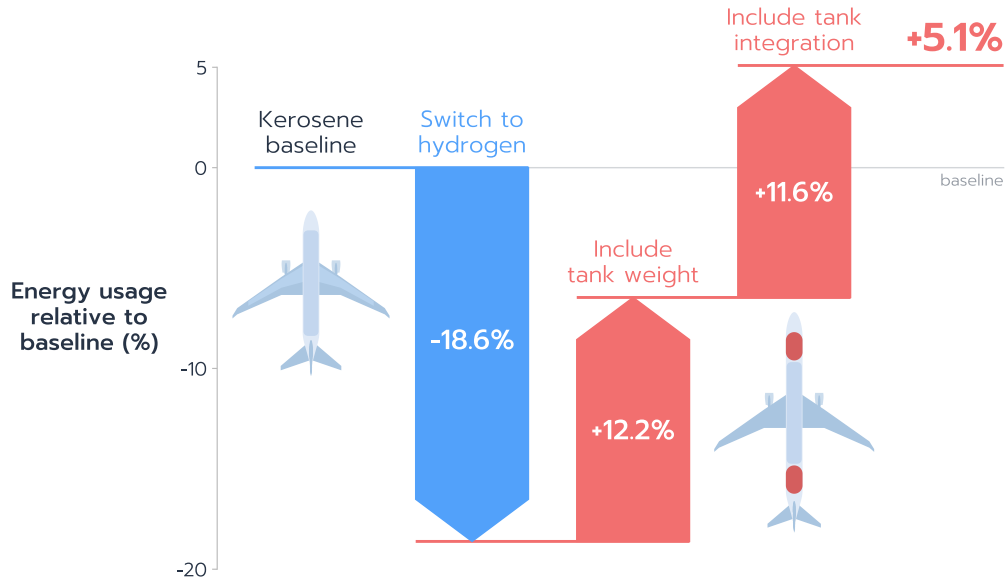


Figure 14: The reduction in fuel weight from hydrogen's greater specific energy than kerosene is negated by the tank weight and integration penalties. This results in a net energy consumption increase of 5.1% for the hydrogen-powered T&W aircraft compared to the kerosene-powered baseline.

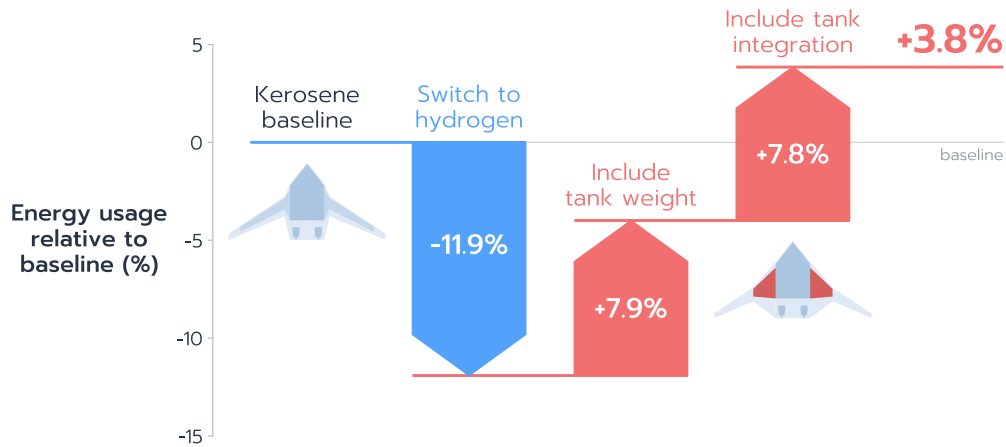


Figure 15: The BWB benefits less from switching to hydrogen than the T&W because of its lower fuel fraction. After including the tank weight and integration penalties, the hydrogen-powered BWB uses 3.8% more energy than the kerosene-powered baseline.

Switching the T&W concept to hydrogen reduces energy consumption by 18.6%, while the BWB saves only 11.9% energy. This difference is due to a combination of factors. One cause is that the BWB concepts have a lower fuel fraction than the T&W aircraft, which is an effect of the BWB's greater aerodynamic efficiency. The lower fuel fraction means that the BWB's takeoff weight decreases less

than the T&W's when the fuel weight is reduced by a factor of three. Another source of the difference is that the wing is a greater fraction of the total weight for the BWB than the T&W. This means the 20% wing weight penalty from removing fuel load alleviation hurts the BWB more than the T&W. A third cause is the indirect effect of switching to hydrogen on the T&W's tail sizing. Switching to hydrogen reduces the takeoff weight. The lower takeoff weight allows the wing to shrink. A smaller wing and lighter aircraft reduce the required size of the horizontal and vertical stabilizers. The horizontal and vertical stabilizers account for 6% of the kerosene T&W's empty weight and 9% of the drag in cruise, so reducing their size can have a significant effect on fuel burn. Because the BWB configuration has no tail, it misses these savings.

Adding the tank weight reduces the benefit of hydrogen for the T&W aircraft by two-thirds. The 8% penalty in energy consumption from adding the tanks to the BWB is smaller than the T&W's. However, the ratio of the tank weight penalty to the initial energy savings from switching to hydrogen is similar for the two configurations. The smaller percent increase may be partly due to the higher aerodynamic efficiency, which results in smaller changes to fuel burn for a given change in weight and lighter tanks due to the lower fuel fraction.

Finally, integrating the hydrogen tanks into the T&W aircraft results in a net increase in energy consumption of 5.1% for the hydrogen-powered version compared to the kerosene-powered baseline. Lengthening the fuselage shrinks the horizontal and vertical tails, which reduces their weight and drag. On the other hand, the longer fuselage increases the fuselage's weight and drag. It also requires longer landing gear to maintain suitable rotation angles for takeoff and landing without striking the tail on the ground, which increases the landing gear weight. Adding the tank fill level constraint to the BWB increases the outerwing planform area by 33%, adding substantial wetted area that increases drag. This yields an energy increase of 3.8% compared to the kerosene BWB baseline.

For safety reasons, it may be required to store the liquid hydrogen tanks outside of the pressurized passenger cabin. The T&W would require two extra pressure bulkheads in the fuselage to accomplish this—one behind and one in front of the forward hydrogen tank. Assuming the two bulkheads add a total of 1,000 kg to the aircraft's weight, the new energy usage difference between the kerosene and hydrogen T&W concepts rises from 5.1% to 6.0%. This modification is speculative and meant to give only a rough idea of the potential impact of extra pressure bulkheads.

These results suggest that the BWB configuration may not be any better at adapting to hydrogen than the conventional T&W configuration. It is possible that higher fidelity aerodynamic and structural simulations would result in a larger blending region on the baseline kerosene BWB. This might increase the baseline BWB's energy consumption and reduce the resulting energy penalty of switching to hydrogen. Either way, the conclusions in this study do not rule out a hydrogen-powered BWB. If the kerosene BWB baseline uses, for example, 50% less energy than the kerosene T&W, a hydrogen BWB still has the potential to use far less energy than a hydrogen T&W. Additionally, the models are only a guess of reality, so they are subject to uncertainty. We investigate the effects of this uncertainty in the following section.

4 Sensitivity Analysis

This section investigates how the conclusions of this paper are affected by modeling assumptions. We explore the sensitivity of the 5.1% and 3.8% numbers from the previous section to the hydrogen tank weight, penalty on the wing weight to account for the lack of fuel load alleviation, hydrogen engine fuel consumption, operating empty weight, and total drag. We also study how the results change when the BWB's conformal tank is extended to fill the entire blending region.

The conclusions in the previous section are based on the energy of each configuration's hydrogen

version relative to its kerosene version:

$$\text{Relative energy usage} = \frac{\text{Hydrogen} - \text{Kerosene}}{\text{Kerosene}} - 1 \quad (5)$$

We measure this metric's sensitivity to the variables of interest. The variables of interest can be divided into two types. The first type affects only the hydrogen concept, so the kerosene baseline's energy usage is unchanged. In the first category are sensitivities to hydrogen tank weight, penalty on the wing weight to account for the lack of fuel load alleviation, and hydrogen engine fuel consumption. The second type affects both kerosene and hydrogen concepts, so both the kerosene and hydrogen concepts' energy usage are recomputed with the new parameter. Sensitivities to operating empty weight and total drag are in the second category because changing these values affects the kerosene and hydrogen results. Interpreting the second type requires more careful thought. Relative energy usage being insensitive to empty weight changes means that the hydrogen and kerosene energy usage are changing at the same rate, not that the energy usage itself is constant.

To examine the sensitivity of the results to the hydrogen tank weight (Figure 16), we apply multipliers to the tank weight estimates for the hydrogen version of each configuration. The resulting energy consumptions are compared to the kerosene-powered baselines. The point where the hydrogen and kerosene versions have similar energy consumptions is 65–70% for both configurations, though the T&W's crossover point is slightly lower. The BWB's greater lift-to-drag ratio means that a weight increase adds less drag than it would for the T&W. Therefore, energy consumption is also less sensitive to weight changes and, thus, to gravimetric efficiency. This flattens the BWB's curve compared to the T&W's. However, the conformal tanks on the BWB likely have a lower gravimetric efficiency than the T&W's tanks. As discussed in Section 2.2.2, this is because their shape does not contain pressure loads as well as the T&W's cylindrical tanks. This effect makes it more challenging for the BWB to compete with the T&W because the hydrogen BWB would fall further to the left on Figure 16 than the T&W would. Based on this sensitivity study, the conclusions are reasonably robust to the tank weight model. If both configurations used tanks with substantially lower gravimetric efficiencies, the BWB configuration may have an advantage.

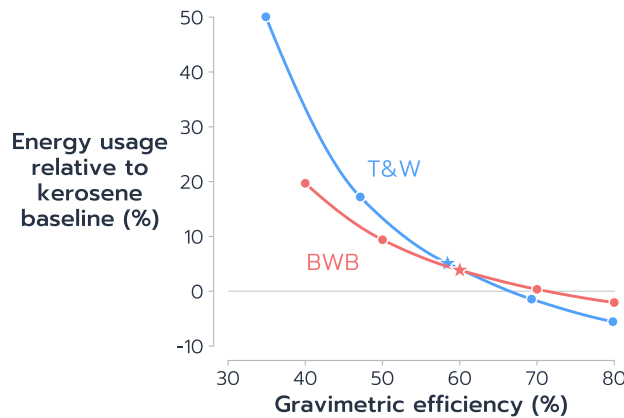


Figure 16: The BWB's higher lift-to-drag ratio makes it less sensitive to weight changes than the T&W. The BWB's lower fuel fraction also means that changing the tank weight has a smaller effect on the aircraft weight. The star marks the gravimetric efficiency assumed in Section 3.

Figure 17 shows the sensitivity of the results to the multiplier on the wing weight that accounts for the lack of fuel load alleviation in the hydrogen concepts' wings. While the default value of 20% is likely reasonable for the T&W's empty wing, it is more uncertain for the BWB concept. The BWB likely falls somewhere between a penalty of 0% and 20%. The sensitivity plot shows that while a

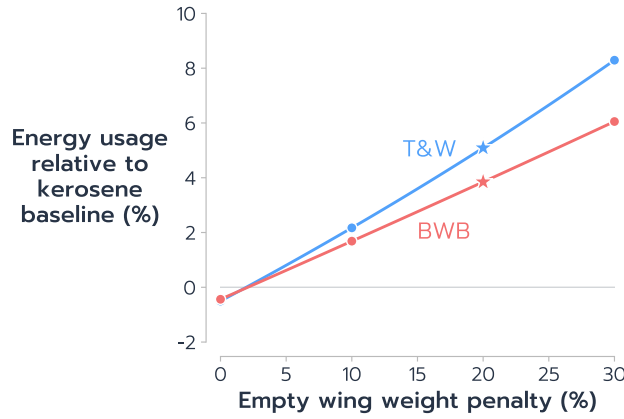


Figure 17: Halving the wing weight penalty that accounts for the lack of fuel load alleviation reduces the BWB’s energy usage by 2%. The star marks the value assumed in Section 3.

smaller wing weight penalty does reduce the energy penalty of switching to hydrogen, it does not dramatically change the conclusions of the study. A wing weight penalty for the BWB of 10% instead of the default 20% reduces the energy increase when switching to hydrogen by 2%. With no wing weight penalty, the T&W and BWB have equal energy penalties for switching to hydrogen. The shallower slope in the sensitivity for the BWB is attributable to the BWB’s greater lift-to-drag ratio, making it less sensitive to weight changes. The results are less sensitive to the wing weight penalty than gravimetric efficiency.

As discussed in Section 2.1.4, taking advantage of hydrogen’s properties in the engine cycle may enable engine efficiency improvements. Figure 18 demonstrates how these efficiency improvements would affect the energy difference between kerosene and hydrogen versions of the two configurations. The response to changes in engine efficiency is roughly linear within the explored range. The T&W and BWB sensitivities follow similar trends, so this study’s conclusions are robust to variations in the hydrogen turbofan’s efficiency. The BWB’s slightly lower sensitivity is caused by its greater lift-to-drag ratio, which makes fuel burn less sensitive to changes in engine efficiency—this relationship can be derived from the Bréguet range equation.

To investigate the sensitivity to empty weight and drag, we apply a multiplier on the total empty weight and drag estimates for both the kerosene baseline and hydrogen aircraft. We compute the

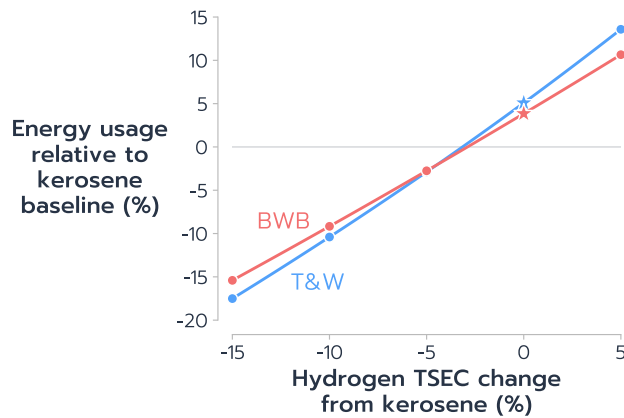


Figure 18: Hydrogen may enable engine efficiency improvements. However, those improvements do not change the conclusions of this study because the T&W and BWB respond similarly to changes in hydrogen engine efficiency. The star marks the value assumed in Section 3.

difference in energy consumption between the modified kerosene and hydrogen aircraft (Equation 5) as we vary the multiplier's value. The results are shown in Figure 19. The difference in energy usage between the hydrogen and kerosene BWBs is nearly constant as the weight and drag models change. This means that changes to the models have similar effects on the energy consumption of the kerosene and hydrogen versions of the BWB (it does not mean that the BWB energy consumptions are unaffected by increased weight). The T&W exhibits a different trend. As weight and drag increase, the hydrogen T&W's energy usage increases relative to the kerosene T&W's. The T&W is particularly sensitive to empty weight.

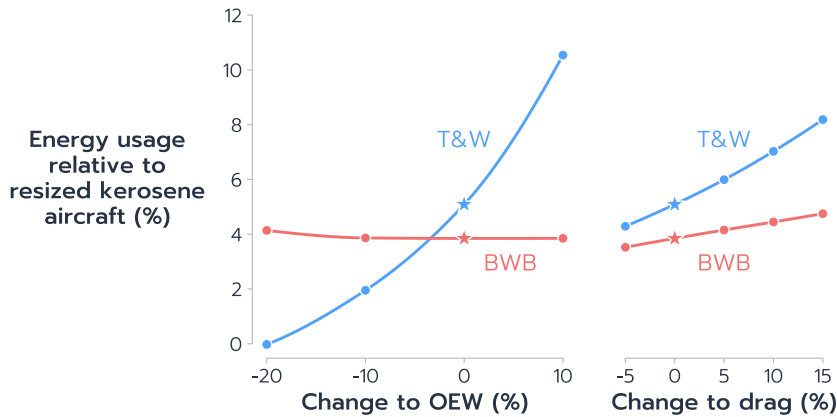


Figure 19: The hydrogen and kerosene BWBs respond similarly to changes in weight and drag, resulting in low sensitivity to these parameters. The star marks the weight and drag assumed in Section 3.

The two configurations' dramatically different sensitivities to empty weight and drag are likely due to how each configuration responds to changes in fuel volume. When the T&W's empty weight increases, the wing area must also increase, adding weight. For the kerosene concept, the larger wing (and tail) is the only geometry change. However, for the hydrogen concept, the hydrogen tanks and fuselage must be lengthened to store the fuel required to carry the additional weight. The longer fuselage adds both drag and weight, and it does not substantially contribute to the additional lift required. This difference between the hydrogen and kerosene T&Ws might be what causes the hydrogen T&W to become less competitive with the kerosene version as the empty weight model is made more conservative. On the other hand, the BWB's wing area increases as the hydrogen tank expands because the tank is contained in a lifting surface. Therefore, increasing the BWB's empty weight increases the wing area of both the kerosene and hydrogen versions. The hydrogen BWB's larger wing can already contain more hydrogen fuel, so the BWB does not experience the same penalty from extending the fuselage as the T&W does. The similar increases in wing area and weight likely explain the consistent energy consumption difference between the hydrogen and kerosene BWBs.

Finally, we extend the length of the conformal tanks to the full chord. This is the best-case scenario for the BWB if all the fuel is stored in the blending regions. Figure 20 shows the optimized result with this tank configuration. Despite the noticeably smaller blending regions, this hydrogen BWB still uses 1.6% more energy than the kerosene BWB. This is likely due to how the conformal tank volume is computed (Figure 7). The tank's cross-sectional areas at the inboard and outboard sections are computed by integrating the area of a supercritical airfoil at the tank's chordwise positions. Most of the airfoil's area is in the first half of the chord. Thus, even though this modification nearly doubles the chord of the tank, the airfoil area taken up by the tank increases only 33%. The result is only a limited improvement in the hydrogen BWB's energy consumption relative to the kerosene BWB baseline.

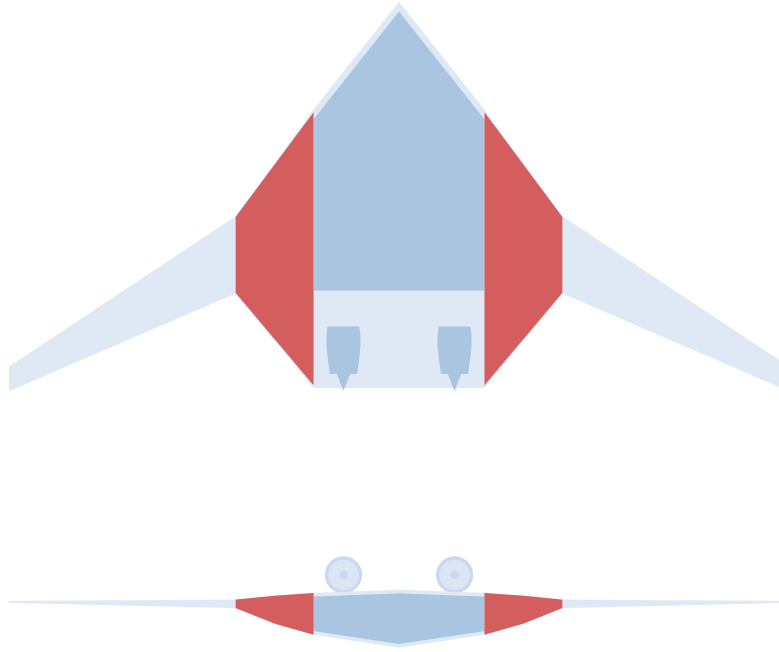


Figure 20: Even when the BWB’s conformal tank is stretched to fill the entire blending region, it still uses 1.6% more energy than the kerosene BWB.

5 Conclusion

We investigate the argument that the BWB configuration naturally lends itself to being powered by hydrogen because it can efficiently store large volumes of hydrogen fuel. To do this, we design kerosene and hydrogen versions of conventional T&W and BWB configurations. We determine how well each configuration adapts to hydrogen by comparing the energy consumption of the hydrogen version of each configuration to the kerosene version.

The BWB configuration is marginally better at adapting to hydrogen fuel than the conventional T&W configuration. The hydrogen BWB uses 3.8% more energy than the kerosene BWB baseline, even with ambitious conformal hydrogen tank gravimetric efficiency assumptions. The hydrogen T&W consumes 5.1% more energy than its kerosene T&W baseline. Hydrogen delivers the same energy as kerosene but with one-third the weight. The kerosene BWB has a lower fuel fraction than the kerosene T&W, so the BWB does not benefit as much as the T&W from hydrogen’s high energy per mass. The energy consumption penalty from integrating hydrogen tanks into the BWB is similar to the T&W’s penalty relative to the amount each configuration benefits from hydrogen’s high energy per mass.

We evaluate the robustness of these conclusions to uncertainty in the tank weight, empty weight, and drag. The BWB results are less sensitive than the T&W’s to all the variables we test. This is partially due to the BWB’s greater lift-to-drag ratio because it results in smaller thrust increases (and thus energy consumption) for a given increase in weight. It is also due to the BWB’s integration of the tanks into a lifting surface. When additional fuel is required, the T&W configuration’s fuselage must be extended to make room for more tank volume, which adds drag and weight. The size of the wing must then increase to support the extra weight, which adds further weight and drag. This requires more fuel, leading to a vicious cycle. Increasing the blending region size of the BWB to contain more fuel already corresponds to an increase in wing area, so the net increase in energy consumption is not as high as for the T&W.

More work is needed to understand the tradeoffs between the T&W and BWB configurations

for hydrogen-powered aircraft. While the hydrogen storage location in the T&W is straightforward, the BWB offers many tank integration possibilities. Future work should investigate alternative tank packaging solutions, particularly in an aerodynamic shape optimization context, while including the impact of design decisions on aircraft sizing.

Despite the BWB showing a slight advantage, these results suggest that BWBs may not be the silver bullet for hydrogen storage that some expect. Rather, they indicate that BWB and T&W aircraft have similar energy penalties when switching from kerosene to hydrogen. This means that a BWB would be just as appealing when powered by hydrogen as kerosene. As a result, BWB technology maturation can benefit the future of aviation whether or not hydrogen becomes the primary fuel.

Appendix

Parasitic drag buildup

This section describes the drag estimation. Much of this drag estimation is based on a synthesis of parasitic drag prediction methods in the OpenVSP documentation¹⁰. The form factors and interference factors for the components are listed in Table 6.

Table 6: Drag estimation form and interference factors

Component	Form factor	Interference factor	Notes
Wing	$1 + 2.7 t/c + 100(t/c)^4$	1.0	From Torenbeek [40]
Tails	Same as wing	1.05	Interference factor from Raymer [27]
Fuselage	$1 + 2.2/f + 3.8/f^3$	1.0	$f = \frac{\text{fuselage length}}{\text{fuselage diameter}}$, from Torenbeek [40]
Nacelle	1.25	1.2	From Jenkinson et al. [35]

The laminar and turbulent skin friction coefficient equations are the Blasius equation and an explicit fit of Spalding [41], respectively, written as

$$C_{f, \text{laminar}}(Re) = \frac{1.32824}{\sqrt{Re}} \quad (6)$$

$$C_{f, \text{turbulent}}(Re) = \frac{0.523}{\ln^2(0.06 Re)}, \quad (7)$$

where Re is the Reynolds number computed based on each component's characteristic length. Only $C_{f, \text{turbulent}}$ is computed when the boundary layer is fully turbulent. When there is some laminar flow, the total skin friction coefficient is computed as

$$C_f(Re) = C_{f, \text{turbulent}}(Re) + \beta (C_{f, \text{laminar}}(\beta Re) - C_{f, \text{turbulent}}(\beta Re)), \quad (8)$$

where β is the fraction of the total boundary layer that is laminar. The wing and tail surfaces are assumed to have 15% laminar flow and the fuselage and nacelle have 0%.

The additional drag from flaps is computed by interpolating a prediction from Roskam [42, Figure 4.48] for Fowler flaps using the following:

$$\Delta C_{d, \text{airfoil}} = 10^{-5} (57.5 c_{\text{flap}}^2 - 7.45 c_{\text{flap}} + 1.23) \delta^2 \quad (9)$$

$$C_{D, \text{flaps}} = \Delta C_{d, \text{airfoil}} A_{\text{flapped}} \cos \Lambda, \quad (10)$$

¹⁰<https://openvsp.org/wiki/doku.php?id=parasitedrag> (accessed April 24, 2023)

where c_{flap} is the ratio of the flap's chord to the wing's chord (we assume 0.2), δ is the flap extension angle in degrees, A_{flapped} is the ratio of wing area that has trailing edge flaps to total wing area (we assume 0.9), and Λ is the sweep angle in degrees. This approach provides a reasonably good fit of the original curves up to a flap deflection angle of around 30 degrees. An interference factor of 1.25 is applied to the flap drag, as suggested by Roskam [42].

Maximum lift coefficient estimation

The increment to the clean maximum lift coefficient from extending the flaps and slats is found by fitting curves to charts from Roskam [42]. The additional lift for the airfoil section from extending the flaps is computed using Roskam [42, Equation 8.18] as

$$\Delta C_{l \max, \text{base}} = 1 + 2.33 \left(\frac{t}{c} \right) - 77.9 \left(\frac{t}{c} \right)^2 + 1120 \left(\frac{t}{c} \right)^3 - 3430 \left(\frac{t}{c} \right)^4 \quad (11)$$

$$k_1 = 4c_{\text{flap}} \quad (12)$$

$$k_2 = 0.4 + 0.0234\delta - 2.04 \times 10^{-4} \delta^2 \quad (13)$$

$$k_3 = 1.36k_2 - 0.389k_2^2 \quad (14)$$

$$\Delta C_{l \max} = k_1 k_2 k_3 \Delta C_{l \max, \text{base}}, \quad (15)$$

where $\Delta C_{l \max, \text{base}}$, k_1 , k_2 , and k_3 are fits of the Fowler flap lines from Roskam [42, Figures 8.31–8.34]. The thickness-to-chord ratio in the first equation is the ratio, not the percent (no multiplication by 100). The increment to the total wing lift from the flaps is then computed using Roskam [42, Equation 8.29]:

$$\Delta C_{L \max, \text{flaps}} = \Delta C_{l \max} A_{\text{flapped}} (1 - 0.08 \cos^2 \Lambda) \cos^{0.75} \Lambda. \quad (16)$$

Finally, the lift addition from the slats is computed with Roskam [42, Equation 8.30]:

$$\Delta C_{L \max, \text{slats}} = 7.11 c_{\text{slat}} b_{\text{slat}}^2 \cos^2 \Lambda, \quad (17)$$

where c_{slat} is the ratio of the slat's chord to the wing's chord (we assume 0.1), and b_{slat} is the ratio of the slat's span to the wing's span (we assume 0.8). The final takeoff $C_{L \max}$ is computed by adding $\Delta C_{L \max, \text{flaps}}$ and $\Delta C_{L \max, \text{slats}}$ to the clean $C_{L \max}$ estimated with Raymer [27, Equation 12.15].

Optimization results

These tables list the optimization problem results for the kerosene T&W (Table 7), hydrogen T&W (Table 8), kerosene BWB (Table 9), and hydrogen BWB (Table 10). For the BWB optimizations, the blending region outboard spanwise position design variable coordinate system is zero at the lateral symmetry plane. It is increasingly negative as the blending region span increases. The upper bound prevents the blending region's outboard section from encroaching on the centerbody. In addition to the tables listing variables included in the optimization problem, Table 11 lists the average cruise lift-to-drag ratio and a detailed component weight breakdown for each of the configurations.

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Table 7: Kerosene T&W optimization results. Bold values indicate active constraints or design variable bounds.

		Unit	Lower	Upper	Value
minimize	$W_{\text{mission fuel}}$	lb			152,907
by varying	Engine rated thrust	lbf			73,369
	Wing area	ft ²			3,864
	Wing twist	deg			-0.26 (tip), 5.14, 2.51, 0.00 (root)
	MTOW	lb			575,978
subject to	Takeoff field length	ft		9,000	8,772
	Throttle		1.0%	100.0%	min = 6.2%, max = 100.0%
	MTOW - (OEW + W_{payload} + $W_{\text{total fuel}}$)	lb	0		-8.83×10^{-5}

Table 8: Hydrogen T&W optimization results. Bold values indicate active constraints or design variable bounds.

		Unit	Lower	Upper	Value
minimize	$W_{\text{mission fuel}}$	lb			57,580
by varying	Engine rated thrust	lbf			70,520
	Wing area	ft ²			4,006
	Hydrogen tank radius	ft		9.7	9.7
	Hydrogen tank length	ft	0.0		13.8
	Wing twist	deg			-0.29 (tip), 5.09, 2.48, 0.00 (root)
	MTOW	lb			539,826
subject to	Takeoff field length	ft		9,000	7,781
	Throttle		1.0%	100.0%	min = 8.6%, max = 100.0%
	Hydrogen tank fill level		0.00	0.95	9.94×10^{-7}
	MTOW - (OEW + W_{payload} + $W_{\text{total fuel}}$)	lb	0		-3.84×10^{-4}

Table 9: Kerosene BWB optimization results. Bold values indicate active constraints or design variable bounds.

		Unit	Lower	Upper	Value
minimize	$W_{\text{mission fuel}}$	lb			112,585
by varying	Engine rated thrust	lbf			59,977
	Blending region mid thickness-to-chord ratio		11.0%		11.0%
	Blending region outboard thickness-to-chord ratio		11.0%		11.0%
	Blending region outboard chord	ft			27
	Blending region outboard spanwise position	ft		-30	-30
	Blending region outboard streamwise position	ft			46
	Wing span	ft		210	210
	Wing twist	deg			2.34 (tip), 2.67, -0.47, 0.00 (root)
	MTOW	lb			514,995
subject to	Takeoff field length	ft		9,000	9,000
	Throttle		1.0%	100.0%	min = 5.0%, max = 93.4%
	Change in streamwise quarter MAC position	ft	0	0	1.04×10^{-10}
	Furthest outboard wing aspect ratio			10	8.57
	MTOW - (OEW + W_{payload} + $W_{\text{total fuel}}$)	lb	0		-3.91×10^{-5}

Table 10: Hydrogen BWB optimization results. Bold values indicate active constraints or design variable bounds.

		Unit	Lower	Upper	Value
minimize	$W_{\text{mission fuel}}$	lb			41,895
by varying	Engine rated thrust	lbf			55,749
	Blending region mid thickness-to-chord ratio		11.0%		15.0%
	Blending region outboard thickness-to-chord ratio		11.0%		11.0%
	Blending region outboard chord	ft			19
	Blending region outboard spanwise position	ft		-30	-53
	Blending region outboard streamwise position	ft			63
	Wing span	ft		210	210
	Wing twist	deg			4.62 (tip), 3.30, -0.14, 0.00 (root)
	MTOW	lb			469,221
subject to	Takeoff field length	ft		9,000	9,000
	Throttle		1.0%	100.0%	min = 8.2%, max = 95.9%
	Hydrogen tank fill level		0.00	0.95	9.98×10^{-7}
	Change in streamwise quarter MAC position	ft	0	0	-2.39×10^{-9}
	Furthest outboard wing aspect ratio			10	8.49
	MTOW - (OEW + W_{payload} + $W_{\text{total fuel}}$)	lb	0		-2.21×10^{-5}

Table 11: Average lift-to-drag ratio and weight breakdown for the four optimized configurations.

	Kerosene T&W	Hydrogen T&W	Kerosene BWB	Hydrogen BWB
Average cruise L/D	19.7	19.4	24.8	23.3
MTOW (lb)	575,978	539,826	514,995	469,221
Fuel (lb)	174,880	65,990	129,836	48,351
Payload (lb)	116,000	116,000	116,000	116,000
OEW (lb)	285,098	357,836	269,158	304,869
Wing (lb)	71,259	84,747	73,518	82,167
Centerbody (lb)	—	—	88,319	86,924
Fuselage (lb)	52,198	61,560	—	—
Propulsion (lb)	50,094	48,150	34,058	31,660
Landing gear (lb)	36,591	44,657	24,958	25,390
Empennage (lb)	17,627	14,652	—	—
Hydrogen tanks (lb)	—	46,956	—	32,234
Other (lb)	57,330	57,113	48,304	46,494

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